

INDUSTRIAL WATER & WASTEWATER ENGINEERING SERIES

ZERO LIQUID DISCHARGE (ZLD) SYSTEM

Detailed Engineering & Operations Manual — Expanded Edition

DETAILED ENGINEERING & OPERATIONS MANUAL — EXPANDED EDITION

Water Chemistry Deep-Dives · Process Flow Diagrams · Design, O&M and Project Reference

Industrial Water & Wastewater Engineering Series · Reference Document for Design, O&M and Project Teams

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1. System Profiling & Overview

Zero Liquid Discharge (ZLD) is a treatment strategy where no liquid effluent leaves the plant boundary – all wastewater is treated, concentrated, and ultimately converted to solid salts/sludge and recovered water, achieving 100% water recycling. ZLD is mandated by regulators for highly polluting/water-stressed sectors such as textile dyeing, tannery, distillery, pulp & paper, and pharmaceutical bulk drug manufacturing in many jurisdictions.

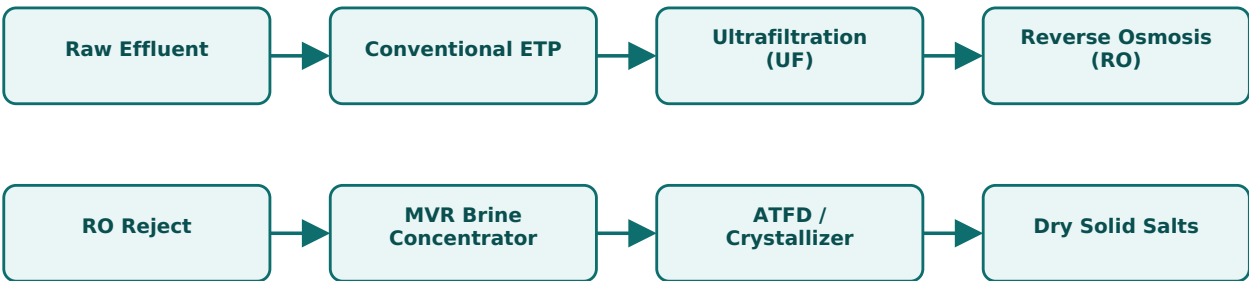


Fig. 1 — General ZLD Process Train, with distillate/permeate recycled to process water

1.1 Drivers for ZLD Adoption

- Regulatory mandate by CPCB/SPCB for red-category and water-intensive industries.
- Water scarcity in industrial clusters driving voluntary maximum water reuse.
- Elimination of discharge-related environmental liability and community disputes.
- Recovery of high-value salts (e.g., Glauber's salt/sodium sulfate, common salt) as a by-product revenue stream in textile and chemical sectors.

2. Feed Stream Characterization & Design Basis

Parameter	Typical ETP-Treated Feed to ZLD	Design Relevance
TDS	3,000-15,000 mg/L	Determines RO staging and brine concentrator sizing
COD (residual)	<250 mg/L (post-ETP)	High residual COD fouls RO membranes; may need AOP polishing
Hardness	Variable, often high in textile effluent	Requires softening before RO to prevent scaling
Silica	10-50 mg/L	Limits achievable RO recovery; critical for brine concentrator scaling control
Sulfate/Chloride ratio	Sector-specific (e.g., high sulfate in textile)	Determines salt recovery quality (mixed salt vs. segregated recovery)

3. Process Design & Technology Train

- **Ultrafiltration (UF):** Removes residual suspended solids and colloids ahead of RO, protecting membranes from particulate fouling; typically hollow-fiber, 0.01-0.05 micron rating.
- **Reverse Osmosis (RO):** Primary desalination step recovering 60-75% of feed as permeate (reused as process water); reject stream (25-40% of feed, now concentrated 3-4x) proceeds to brine management.
- **High-Recovery RO / Disc Tube Module (DTRO):** Handles higher TDS reject streams tolerant of scaling/fouling, pushing overall system recovery higher before thermal concentration.
- **Mechanical Vapor Recompression (MVR) Brine Concentrator:** Evaporates RO reject water using energy-efficient vapor recompression, concentrating brine to near-saturation (10-25% TDS), recovering distillate as high-purity condensate.
- **Agitated Thin Film Dryer (ATFD) / Crystallizer:** Final stage evaporates remaining water from concentrated brine to produce dry solid salt cake, achieving true zero liquid discharge.

4. Pretreatment Systems

Stage	Purpose	Typical Equipment
Conventional ETP (upstream)	Reduce organic load and suspended solids before membrane stages	Biological + physico-chemical ETP (see ETP manual)
Softening	Prevent hardness scaling on RO/brine concentrator	Ion-exchange softener, sized for high-TDS duty
Ultrafiltration	Remove residual colloidal/suspended solids	Hollow-fiber UF skid
Antiscalant & pH Dosing	Control scaling potential on RO membranes	Dosing system with antiscalant, acid for pH adjustment
Cartridge Filtration	Final fine-particle polishing before RO	5 micron cartridge filter

5. Concentration & Crystallization Stages

The economics and reliability of ZLD hinge on maximizing membrane-stage recovery (lower cost per m3) before switching to thermal evaporation (higher cost per m3 but able to handle very high TDS). A well-designed ZLD system pushes RO/DTRO recovery to 85-95% of total feed, so the brine concentrator and crystallizer – the most capital and energy-intensive stages – only need to process the remaining 5-15% reject volume.

- Seeded slurry crystallizers or forced circulation crystallizers are used where salt purity for resale is a design objective (e.g., segregated sodium sulfate/sodium chloride recovery in textile ZLD).
- Falling Film Evaporators are typically staged ahead of ATFD to reduce load on the final drying stage.
- Condensate/distillate from evaporation stages is polished (activated carbon, ion exchange if needed) and returned to the process water tank.

6. Brine & Crystallization Chemistry Deep-Dive

6.1 Solubility-Driven Staged Concentration

Solubility (g/100mL water, ~25°C, approx.):
CaSO₄ (gypsum): ~0.2 – precipitates FIRST as brine concentrates
CaCO₃: ~0.0015 – precipitates very early, must be removed by softening
Na₂SO₄ (Glauber's salt): ~28 (highly temperature dependent)
NaCl: ~36 – stays in solution longest, crystallizes LAST

Because different salts have very different, and in some cases sharply temperature-dependent, solubility limits, a ZLD brine concentrating in stages will naturally precipitate low-solubility salts (calcium sulfate, calcium carbonate, silica) first – these are the species that scale MVR heat-transfer tubes (Section 7) – while highly soluble salts (sodium chloride, sodium sulfate) remain in solution until the final crystallizer stage. Segregated salt recovery schemes exploit this solubility gap, using controlled cooling/seeding to crystallize sodium sulfate selectively before sending the remaining liquor to a separate NaCl crystallizer.

6.2 Scaling Control in the MVR Brine Concentrator

Seeded slurry approach: maintain a suspended seed-crystal population so that supersaturated calcium sulfate/silica preferentially deposits on existing seed crystals (bulk crystallization) rather than nucleating on the hot heat-exchanger tube wall (surface scaling)

This is the same principle used in seeded softening/lime clarifiers, applied at high-TDS conditions: keeping the bulk liquor supersaturated but the seed crystal surface area high enough diverts crystal growth away from the heat transfer surfaces, which is the main lever for controlling MVR fouling rate alongside periodic acid cleaning.

6.3 Mass & Salt Balance (worked example)

Feed to ZLD = 100 m³/day at TDS 8,000 mg/L → Salt load = 800 kg/day
RO/DTRO recovery = 90% → Permeate = 90 m³/day (reused); Reject = 10 m³/day, now carrying all 800 kg TDS → concentrate TDS ≈ 80,000 mg/L
MVR concentrates further to 20% w/w brine → final crystallizer feed volume ≈ 4 m³/day → dry salt cake output ≈ 800 kg/day (minus any moisture in cake)

This mass-balance approach (tracking total dissolved salt mass rather than volume alone) is the standard way to verify a ZLD system's actual recovery/salt-balance performance during a PGTR and to size the crystallizer and salt handling/storage system correctly.

6.4 Distillate/Condensate Purity

Because evaporation is a phase-change (liquid to vapor) process, MVR/ATFD distillate is inherently very low in TDS (typically <10-50 mg/L) regardless of the feed brine concentration – the main quality risk is mechanical carryover (droplet entrainment) of un-evaporated brine through a fouled or poorly designed demister, not a chemistry limitation. This is why distillate TDS is a good diagnostic for demister condition (Section 9/10) rather than for the evaporator's thermal performance.

7. Common Problems & Engineering Solutions

Problem	Root Cause	Engineering Solution
RO membrane scaling/fouling	Inadequate softening, poor antiscalant dosing, high silica	Improve upstream softening, optimize antiscalant per feed chemistry, reduce recovery if silica-limited
Brine concentrator scaling on heat transfer tubes	Calcium sulfate/silica precipitation at high concentration	Seed crystallization control, periodic acid cleaning, review feed pretreatment
ATFD frequent choking/fouling	High viscosity slurry, uneven wiper blade wear, poor temperature control	Inspect/replace wiper blades, verify feed concentration within design range
Poor salt purity for resale	Mixed salt streams not segregated, inadequate crystallization control	Consider segregated recovery train (softening + selective crystallization) if resale value justifies capex
High energy consumption	Sub-optimal MVR compressor operation, poor	Energy audit, verify MVR compressor efficiency, review heat

	heat integration	exchanger fouling
Low overall system recovery (below design)	Conservative RO staging, membrane fouling, poor DTRO utilization	Review RO staging/DTRO addition, membrane cleaning (CIP) frequency
Odor/corrosion in brine handling area	Concentrated chloride/sulfate corrosivity, H2S generation in stagnant zones	Use corrosion-resistant MOC (SS316L/duplex/FRP-lined), improve ventilation

8. Equipment Service Life

Equipment	Typical Service Life	Key Life-limiting Factors
UF Membranes	5-7 years	Fouling, chemical cleaning frequency, fiber breakage
RO/DTRO Membranes	3-5 years (RO); 2-4 years (DTRO, high fouling duty)	Scaling, fouling, oxidant damage
MVR Compressor	15-20 years (major overhaul every 5-7 yrs)	Bearing wear, blade erosion, vibration
Brine Concentrator Heat Exchangers	10-15 years	Scaling, corrosion, thermal cycling stress
ATFD Wiper Blades/Rotor	1-2 years (blades); 10+ years (rotor/shell)	Abrasive wear from salt slurry
Crystallizer Vessel (SS316L/Duplex)	15-20 years	Corrosion, erosion at high-velocity zones
Softener Resin (upstream)	3-5 years	High-TDS duty accelerates fouling
Dosing Pumps	5-7 years	Chemical compatibility, diaphragm wear

9. Preventive Service & Maintenance Schedule

Frequency	Activity
Daily	Monitor RO/DTRO permeate flow and quality; check MVR compressor parameters; verify ATFD product moisture; log salt cake output
Weekly	CIP evaluation based on differential pressure trend across UF/RO; inspect dosing systems; check crystallizer slurry density
Monthly	Full membrane performance normalization (flux/rejection); inspect ATFD wiper blades; lab test salt purity
Quarterly	Heat exchanger inspection for scaling; vibration analysis on MVR compressor; softener resin capacity check
Annual	Complete shutdown inspection of crystallizer/evaporator vessels; MVR compressor major service; membrane replacement planning; statutory ZLD compliance audit

10. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
RO permeate flow declining	Membrane fouling/scaling	Perform CIP, check antiscalant dosing accuracy
MVR compressor tripping on high vibration	Blade fouling/erosion, bearing wear	Inspect and clean/replace blades, check bearing condition
ATFD not achieving dry solid output	Feed too dilute, heating medium temperature low, wiper wear	Verify feed concentration from upstream evaporator, check heating steam/thermic fluid parameters
Excess condensate/distillate quality poor (high TDS)	Entrainment/carryover from evaporator, demister fouling	Inspect and clean demister pads, check operating vacuum/level control
Salt cake sticky/not free-flowing	Incomplete crystallization, high residual moisture	Adjust ATFD temperature/residence time, review crystallizer seed control
Sudden increase in reject volume	RO/DTRO fouling reducing recovery	Investigate membrane condition, review CIP effectiveness

11. Project Execution & Engineering Workflow

Phase	Key Deliverables / Activities
Feasibility & Mass Balance	Detailed feed characterization, technology train selection, water/salt mass balance, capex-opex estimate.

Basic Engineering	PFD across all stages (ETP-UF-RO-MVR-ATFD), equipment sizing, salt recovery study, energy balance.
Detailed Engineering	P&IDs for each stage, GA drawings, metallurgy selection (corrosion-resistant MOC), electrical/control philosophy, HAZOP.
Statutory Approvals	SPCB approval of ZLD scheme, hazardous waste (salt) disposal authorization.
Procurement	Membranes, MVR compressor, ATFD/crystallizer, softener, dosing systems, instrumentation – long-lead items need early ordering.
Civil & Structural	Foundations for heavy rotating equipment (MVR), tankage, salt storage/handling shed.
MEI Erection	Equipment installation and alignment, corrosion-resistant piping, cabling, SCADA/PLC integration.
Pre-commissioning	Hydro test, membrane wet-out and integrity test, dry-run of rotating equipment.
Commissioning	Sequential stage-wise commissioning (UF → RO → MVR → ATFD), ramp-up to design recovery.
Performance Guarantee Test Run	Sustained run (commonly 15-30 days) verifying recovery %, salt purity and zero-discharge compliance.
Handover & O&M	O&M manual, salt disposal/sale arrangement documentation, operator training, spares philosophy.

12. Statutory Compliance & Performance Guarantee

- SPCB/CPCB ZLD scheme approval typically requires demonstrating 100% water recycle with documented salt disposal/sale route (secured landfill for hazardous salts, or sale for non-hazardous industrial salts such as sodium sulfate).
- Continuous online monitoring (flow meters at every recycle/reject point) commonly mandated for regulatory verification of zero discharge.
- Performance guarantee typically covers: overall water recovery % (e.g., $\geq 95-98\%$), permeate/product water quality, salt cake moisture content, and specific energy consumption (kWh/m³ of feed processed).
- Guarantee demonstrated over an extended continuous run (commonly 15-30 days) since ZLD system stability, especially crystallizer performance, takes longer to establish than conventional ETP.

Technical note: figures in this manual are general design guidelines. Final technology train, membrane staging, MVR/ATFD sizing and metallurgy must be established from actual feed characterization and pilot trial data.